EUMETSAT OSI SAF
OCEAN AND SEA ICE

PRODUCT USER MANUAL

For Reprocessed Sea Ice Concentration from EUMETSAT OSI SAF

SEAICE_GLO_SEAICE_L4_REP_OBSERVATIONS_011_009

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2.8	2020.11.09		Added links to OSI SAF documents	C. Wettre	C. Derval
2.9	2021.08.24		Updated links to OSI SAF webportal	S Eastwood	C Wettre
2.10	2023.05.31		New version v3 from OSI SAF	S Eastwood	C Wettre
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TABLE OF CONTENTS

I INTRODUCTION *7*
I.1 Description summary of Products covered by this document **7**
I.2 Product licensed **7**
I.3 Document organisation **7**
II ALGORITHM DESCRIPTION *9*
III PRODUCT DESCRIPTION *10*
IV FILES NOMENCLATURE AND FORMAT *12*
IV.1 Nomenclature of files when downloaded through the Subsetter Service **12**
IV.2 Nomenclature of files when downloaded through the Copernicus Marine FTP Service **12**
IV.3 Land mask and missing values **13**
IV.4 File Format **13**
IV.5 Structure of files **13**
V REFERENCES *14*

GLOSSARY AND ABBREVIATIONS

Calval	Calibration Validation
MIS	Marine Information System
DU	Dissemination Unit. A Dissemination Unit is a kind of Production Unit, with the ability to disseminate its own products or other Production Unit's products to the rest of CMEMS Production Centres and to external Users.
EC	European Commission
DT	Delayed Time
ECMWF	European Centre for Medium Range Weather Forecast
ECV	Essential Climate Variables
EO	Earth Observation
ESA	European Space Agency
EU	European Union
EUMETSAT	European Meteorological Satellite agency
GCOS	Global Climate Observing System
GLO	Global Monitoring and Forecasting Centre
GMES	Global Monitoring for Environment and Security
GTS	Meteorological data exchange network
I/F	Interfaces
L1	Level 1 data : Raw data (telemetry) from instruments
L2	Level 2 data : (A and B) ground segment geophysical measures, interpreted, calibrated
L3	Level 1 data : Multi sensor data
L4	Level 4 data : Analysed data

MCS	Marine Core Service
MFC	Monitoring and Forecasting Centre
NetCDF	Network Common Data Form
NOAA	National Oceanic and Atmospheric Administration of the USA
NRT	Near Real Time
NWP	Numerical Weather Prediction
SI TAC	Sea Ice Thematic Assembly Centre
PUM	Product User Manual
QUID	Quality Information Document
R&D	Research and Development
REA	Reanalysis (for Models)
REP	Reprocessing (for Observations)
SAF	EUMETSAT Satellite Application Facility
SLA	Service Level Agreement
SUBS	Subsetter Download mechanism : CMEMS service tool to download a NetCDF file of a selected geographical box using values of longitude an latitude, and time range
TAC	Thematic Assembly Centres
WMS	Web Map Service

I INTRODUCTION

I.1 Description summary of Products covered by this document

This guide describes the global reprocessed sea ice concentration product SEAICE_GLO_SEAICE_REP_OBSERVATIONS_011_009 from the Copernicus Marine Service Sea Ice Thematic Assembly Centre (Sea Ice TAC), which is a redistribution of the EUMETSAT OSI SAF Climate Data Records (CDR) and Interim Climate data Records (ICDR) and Interim Climate Data Record (ICDR), labelled OSI-450-a1, OSI-458, OSI-430-a and OSI-438.

The guide describes what data services are available to access them, and how to use the files and services. It contains two sets of CDR/ICDRs of reprocessed sea ice concentration (0-100%):

1. CDR (OSI-458) and ICDR (OSI-438) based on AMSR-E and AMSR2 data. This CDR covers the period from 01.06.2002 to 31.12.2020. The ICDR covers the period from 01-01-2021 to present, with 16 days latency.
2. CDR (OSI-450-a1) and ICDR (OSI-430-a) based on SMMR, SSM/I and SSMIS data. This CDR covers the period from 25-10-1978 to 31-12-2020. The ICDR covers the period from 01-01-2021 to 30-09-2025.

For services that rely on updated ICDR products, users are advised to use OSI-438, since the production of OSI-430-a has stopped, due to lack of available input data.

The data are freely available under the Copernicus Marine Service data license. Detailed information on the systems and products is on the Copernicus Marine Service web site: https://marine.copernicus.eu.

I.2 Product licensed

The OSI SAF products are licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0) by EUMETSAT. Whether you're displaying the product as-is or if you have used it as input data, please use the attribution suggested on OSI SAF usage policy and acknowledgement on https://osi-saf.eumetsat.int.

I.3 Document organisation

This Copernicus Marine Service Sea Ice TAC product has been processed by the EUMETSAT OSI SAF. It is documented in the Product User Manual for OSI-458, OSI-438, OSI-450-a1 and OSI-430-a version 3.3, maintained by the OSI SAF project, which can be accessed at any time from the OSI SAF web site at:

https://osisaf-hl.met.no/sites/osisaf-hl/files/user_manuals/osisaf_cdop3_ss2_pum_sea-ice-conc-climate-data-record_v3p3.pdf [RD-1]

For access to documentation on the AMSR-based sea ice concentration CDR and ICDRs:

https://osi-saf.eumetsat.int/products/osi-458 (CDR)
https://osi-saf.eumetsat.int/products/osi-438 (ICDR)

and the SSMIS-based sea ice concentration CDR and ICDRs:

https://osi-saf.eumetsat.int/products/osi-450-a1 (CDR)
https://osi-saf.eumetsat.int/products/osi-430-a (ICDR)

The following chapters will refer explicitly to the corresponding parts of the OSI SAF document, by reference [RD-1].

II ALGORITHM DESCRIPTION

See Section 4 in [RD-1].

III PRODUCT DESCRIPTION

See Section 5 in RD-1.

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

IV FILES NOMENCLATURE AND FORMAT

IV.1 Nomenclature of files when downloaded through the Subsetter Service

SEAICE_GLO_SEAICE_L4_REP_OBSERVATIONS_011_009 files nomenclature when downloaded through the CMEMS Web Portal Subsetter is based on product dataset name and a numerical transaction identifier.

The scheme is: datasetname_nnnnnn.nc

where:

datasetname is a character string within one of the following:

- For the AMSR-based CDR:
  - TBD
  - TBD
- For the AMSR-based ICDR:
  - TBD
  - TBD
- For the SSMIS-based CDR:
  - ice_conc_nh_lamb
  - ice_conc_sh_lamb
- For the SSMIS-based ICDR:
  - ice_conc_nh_crlamb
  - ice_conc_sh_crlamb

. nnnnnn: 6 digit transaction identifier.

.nc: standard NetCDF filename extension.

Example:

ice_conc_nh_lamb_151494.nc

IV.2 Nomenclature of files when downloaded through the Copernicus Marine FTP Service

See Section 5.3 and 5.4 in [RD-1]. The dataset names are as listed in section IV.1.

IV.3 Land mask and missing values

See Section 3.7 and 5.1.2 + 6.1.4 in [RD-1].

IV.4 File Format

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF
To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV.5 Structure of files

See Section 4 in [RD-1].

V REFERENCES

See Section 5 in [RD-1].